MIL-P-37232
20 May 1975

MILITARY SPECIFICATION

POINT, PULP CANAL, DENTAL,

STANDARDIZED GUTTA PERCHA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers standardized gutta percha pulp canal points suitable for use in endodontic treatment procedures.

1.2 Classification. The pulp canal points shall be of the sizes indicated in table I (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6520

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The material shall be high grade Gutta Percha or Balata which shall be washed and treated to remove impurities and provide a uniform, clean, non-toxic product. A filler containing barium sulfate for X-ray opacity shall be uniformly distributed in the finished product. All ingredients shall be of high quality suitable for use in the formulation of pulp canal points.

3.2 Style, design and dimensions. The points shall be designed for use in filling pulp canals in endodontic procedures. The points shall be styled in accordance with the standardized system of the American Association of Endodontists with a standard taper of 0.01875 millimeter (mm) of diameter per mm of point length from D1 to D2 (see table I). The points shall be machine rolled to the required taper and shall have a flattened non-operative end and a rounded or blunt operative end. The flattened non-operative end shall have a serrated or grooved surface pattern which shall be 2 to 4 mm in length and at least one-half the D2 diameter in thickness. The overall length of the points shall be 26 to 32 mm. The points shall be furnished in the sizes and corresponding diameters as shown in table I for the applicable size number specified (see 6.1).

Table I. Diameters of Points

Size No. (Point No) 1/	D 1 (mm) 2/	D 2 (mm) 3/
25	0.25	0.55
30	0.30	0.60
35	0.35	0.65
40	0.40	0.70
45	0.45	0.75
50	0.50	0.80
55	0.55	0.85
60	0.60	0.90
70	0.70	1.00
80	0.80	1.10
90	0.90	1.20
100	1.00	1.30

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1/ Diameter tolerances for sizes 25 through 40 shall be ± 0.10 mm, and for sizes 50 through 100 shall be ± 0.08 mm.

2/ D1 is smallest full diameter at distal tip (disregarding any overhang or concavity at tip resulting from the rolling process).

3/ D2 is diamter at 16 mm from D1.

3.3 Working properties. The gutta-percha points furnished shall be of such formulation and process as to provide a complete sealing of the prepared root canal when used in endodontic procedures. The working properties of the points shall exhibit cohesive strength and toughness (flexibility without brittleness). The points shall be capable of being adapted and condensed in a prepared root canal, forming a dense, dimensionally stable and tissue tolerated filling material. The working properties shall be equal to Mynol Chemical Co.'s "Mynol Gutta Percha Points, AAE Style".

3.4 Color. The color of the points shall be pink, coral, or rose.

3.5 Quantity. Each package shall contain a minimum of 100 points of the size specified.

3.6 Delivery schedule. No more than 2 months shall have elapsed from date of manufacture of the points to date of delivery to the Government.

3.7 Lot or batch number. Each unit package of points shall be marked with the lot or batch number and date of manufacture of the formulation.

3.8 Shelf life. The points shall comply with requirements of this specification for no less than 12 months when stored at 36 to 45° Fahrenheit (F).

3.9 Irritation. The supplier shall furnish a statement directly to the Technical Services Branch (Code ATTS-2), Defense Personnel Support Center, 2800 South 20th St., Philadelphia, Pennsylvania, 19101, stating that the material does not normally irritate oral tissues and does not contain poisonous ingredients in sufficient concentration to be harmful to human beings when used as directed.

3.10 Workmanship. Workmanship shall be first class throughout. Points shall be free of curled or twisted portions. Points and their containers shall be free from defects which detract from their appearance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of compliance. Certificates of compliance, supplied by the manufacturer of the raw materials (gutta percha or Balata and fillers), may be furnished in lieu of actual performance of such testing by the contractor. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, III and IV (see 4.2.1, 4.3, 4.3.1, 4.3.2 and 4.3.3) and 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. The unit of product for examination shall be the individual size point for Major and Minor defects.

Table II. Examination Criteria

Type defect	Inspection level	AQL (Percent defective)
Major	S-4	2.5
Minor	S-1	4.0

4.2.2 Sampling for tests. Sample size for Working Properties test shall be 10 individual size points selected at random from each lot or batch. A lot shall be rejected if any of the samples fail the test.

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4.3 Examination. The points shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with 4.2.1 The points shall be examined for defects, including but not limited to, those listed in table III.

Table III. Classification of Defects

Categories	Defects
Major	
101	Points are not clean.
102	Style and design not as specified.
103	Color of points not as specified.
104	Tapered length surface not smooth rolled finish.
105	Laminations cracks or voids present.
106	Twisted or curled sections present.
Minor	
151	Dimensions not as specified.

4.3.1 Dimensional examination. The diameters D1 and D2 shall be measured by the use of a measuring microscope, shadowgraph, or other suitable device capable of measuring the diameters to 0.01 mm as specified in table I. The lengths shall be measured as above or by use of properly calibrated micrometers, gages or other suitable devices.

4.3.2 Visual examination. Examination for cleanliness, color, shape, uniformity and workmanship shall be made without magnification. A concavity or overhang at the operative end as a result of the machine rolling of the cylindrical stock material shall not be cause for rejection.

4.3.3 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.

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Table IV. Examination for Preparation for Delivery (cont'd)

Examination	Defect
Packaging & Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper location, sequence, or method of application.

4.4 Tests.

4.4.1 Working properties. The gutta-percha points shall be tested for characteristics specified in 3.3. An extracted natural tooth or a tooth die, with a prepared root canal procedure, shall be utilized for evaluating the characteristics specified.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing one hundred points of the size specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. One hundred points of one size only shall be packaged in one or more clear glass or plastic vials of suitable size. Vials shall be fitted friction caps or plug type closures, in addition, vials shall contain cotton or other suitable cushioning material to prevent damage to the points. As an alternate, points may be packaged in a compartmented plastic tray. Each compartment shall be fitted with a snug fitting polyurethane insert to prevent spilling of contents. Vials or trays shall be further packaged in a snug fitting box constructed in accordance with PPP-B-676, Type I. The box shall be wrapped and sealed in a clear plastic wrap.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in a snug fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure of the box shall be adequate to prevent spilling of contents.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Fifty intermediate packages (600 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraph 30.4 of PPP-B-636.

5.2.2 Level B. Fifty intermediate packages (600 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Lot (control) number, contract or purchase order number and name of contractor shall be shown. Type II shelf life marking as specified in MIL-STD-129 shall also be shown. Marking shall include the legend or similar statement:

"STORE BETWEEN 2° - 8°C (35° - 46° F)."

5.3.2 Exterior container. Exterior containers shall be marked as specified in MIL-STD-129. Lot (control) number and the following or similar legend shall be shown:

"STORE BETWEEN 2° - 8° C (35° - 46° F)."

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Type II shelf life marking as specified in MIL-STD-129 shall also be shown.

NOTE: The unrefrigerated shipping time for this item shall be not more than thirty (30) days.

DD Form 1502, 1502-1 or 1502-2 as specified in MIL-STD-129, shall not be required.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size Number (Point No.) required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Supersession data. This specification includes the requirements of Defense Medical Purchase Description No. 1, dated 20 August 1971, for National Stock Number 6520-00-257-3030.

6.3 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-257-3030	POINT, PULP CANAL, DENTAL Standardized, Gutta-Percha, No. 30, 100s
6520-00-257-3031	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 40, 100s
6520-00-257-3032	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 50, 100s
6520-00-257-3033	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 60, 100s
6520-00-257-3036	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 70, 100s

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NSN	Item Identification
6520-00-257-3037	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 80, 100s
6520-00-583-3388	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 25, 100s
6520-00-583-3389	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 90, 100s
6520-00-583-3402	POINT, PULP CANAL, DENTAL Standardized, Gutta-Percha, No. 100, 100s
6520-00-596-7290	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 35, 100s
6520-00-596-7292	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 45, 100s
6520-00-596-7311	POINT, PULP CANAL, DENTAL, Standardized, Gutta-Percha, No. 55, 100s

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1393
is

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-37232 POINT, PULP CANAL, DENTAL, STANDARDIZED GUTTA PERCHA

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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